

Flutter Roadmap

PageView, PageViewBuilder and Responsive

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PageView

PageView is a Flutter widget used to display different pages, the user can move between the pages by swiping left or right.

It is useful for:

- Onboarding screens
- Welcome screens
- Image galleries
- Different screens with the same layout

Important properties:

- children: contains the pages.
- scrollDirection: controls the direction of scrolling.
- onPageChanged: tells us when the user changes the page.
- controller: allows us to control the PageView from the code.

PageView.builder

PageView.builder is used to create pages dynamically, instead of writing every page manually, we use itemBuilder to create the pages.

Important properties:

- itemCount: the number of pages.
- itemBuilder: builds each page.
- index: tells us the number of the current page, it starts from 0.

It is very useful when we have a List of data and want to create pages from this data.

Difference between PageView and PageView.builder

PageView:

- We write the pages directly.
- Good for a small number of fixed pages.
- Simple to use.

PageView.builder:

- Pages are created dynamically.
- Uses itemBuilder and index.
- Good when pages come from a List or the number of pages can change.

For our Onboarding, PageView.builder can be useful because we can store the information of all onboarding pages in a List and build each page using its index.

Responsive Design

Responsive Design means making the UI work correctly on different screen sizes. For example, the application should work well on:

- Small phones
- Large phones
- Tablets
- Different screen resolutions

The UI should adapt to the available screen size instead of depending only on fixed sizes.

Why is it important?

Because different devices have different screen sizes.

Responsive design helps to:

- Avoid overflow problems.
- Make the UI look better.
- Support different devices.
- Give the user a better experience.

